

Executive Summary – Customer Churn Analysis

Dataset Snapshot

- Total Customers Analyzed: **7,043**
- Total Features Evaluated: **21**
- Target Variable: **Customer Churn**
- Churned Customers: **1,869 (26.5%)**
- Retained Customers: **5,174 (73.5%)**

The analysis was conducted using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to identify hidden churn patterns, customer behavior trends, and revenue-impacting factors.

Project Overview

This analysis explores customer churn patterns using a telecom customer dataset containing 7,043 customer records and 21 variables related to demographics, services, billing behavior, contract type, and customer tenure. The objective was to identify the key drivers of customer churn and uncover actionable insights that can help improve customer retention and revenue stability.

Key Findings

1. Overall Churn Rate

- Approximately **26.5% of customers have churned**, while **73.5% of customers remain retained**.
- This means that nearly **1 out of every 4 customers leaves the service**, creating a direct impact on recurring revenue and customer acquisition costs.
- The churn distribution indicates a moderately high retention challenge for the business.
- A lower churn rate is essential for sustainable growth because retaining customers is generally more cost-effective than acquiring new ones.

2. Customer Tenure Strongly Impacts Retention

- Customers with a tenure of **less than 12 months** contribute the majority of churn cases.
- Newly acquired customers, especially those within the **first 1–2 months**, show the highest probability of leaving.
- Long-term customers with tenure above **24 months** demonstrate significantly stronger loyalty and retention.
- The analysis clearly indicates that the **customer onboarding phase is the most vulnerable stage in the customer journey**.
- Businesses that improve first-time user experience, onboarding support, and early engagement campaigns are more likely to reduce churn rates substantially.

3. Contract Type Influences Churn Behavior

- Customers subscribed to **month-to-month contracts** account for the highest share of churned users.
- In comparison, customers using **1-year and 2-year contracts** show significantly lower churn percentages.
- Long-term contract customers are more stable because they are more engaged with the service ecosystem and less likely to switch providers frequently.
- The findings suggest that encouraging customers to move from flexible monthly plans to longer-term plans can improve customer retention and revenue predictability.

4. Senior Citizens Have Higher Churn Risk

- Senior citizens churn at a significantly higher rate (**41.7%**) compared to non-senior customers (**23.6%**).
- This indicates a need for better engagement, support, and service personalization for older customer segments.

5. Service Usage Patterns Affect Churn

- Customers who actively use additional support and security-related services tend to remain more loyal.
- Users without support-oriented services are more likely to churn due to lower perceived value and weaker platform engagement.
- Fiber optic internet customers exhibit higher churn percentages compared to DSL users, potentially indicating service quality or pricing concerns.
- Customers lacking the following services showed noticeably higher churn levels:
 - Tech Support
 - Online Security
 - Online Backup
 - Device Protection

These services appear to increase customer dependency on the platform and improve long-term retention.

- Customers without value-added services such as:
 - Tech Support
 - Online Security
 - Device Protection
- Online Backup are more likely to churn.
- Customers using **Fiber Optic internet services** also show relatively high churn levels.
- This suggests that service quality expectations and support experience strongly influence retention.

6. Payment and Billing Preferences Matter

- Customers using **electronic check payments** have the highest churn rates.
- Customers using **automatic bank transfer or credit card payments** demonstrate stronger retention.
- Users enrolled in **paperless billing** also show noticeably higher churn.

7. Revenue Impact of Churn

- Non-churned customers contribute nearly **five times more revenue** than churned customers.
- Customers with higher tenure generate the majority of total revenue.
- Retaining existing customers is therefore significantly more cost-effective than acquiring new ones.

8. Charges and Churn Relationship

- Customers facing **high monthly charges with low tenure** are more likely to churn.
- Long-term customers remain comparatively stable even with higher total charges.
- This indicates that perceived value and early customer experience are critical factors influencing churn decisions.

Business Recommendations

Based on the analysis, the following strategies can help reduce churn:

1. Strengthen Early Customer Engagement

2. Focus retention campaigns on customers within the first 3–6 months.
3. Improve onboarding and first-time user experience.

4. Promote Long-Term Contracts

5. Encourage annual subscription plans through discounts and loyalty benefits.

6. Improve Support Services

7. Bundle services such as Tech Support and Online Security to increase customer stickiness.

8. Target High-Risk Segments

9. Create dedicated retention strategies for senior citizens and high-bill customers.

10. Optimize Payment Experience

11. Encourage automated payment methods to improve retention.

12. **Reduce Price Sensitivity**

13. Provide personalized offers or loyalty rewards for customers with high monthly charges.

Conclusion

The analysis reveals that customer churn is heavily influenced by tenure, contract structure, support services, payment behavior, and monthly charges. Customers who stay longer, use additional services, and adopt stable payment methods contribute substantially more revenue to the business.

By focusing on early-stage retention, improving customer support offerings, and incentivizing long-term engagement, the company can significantly reduce churn and improve customer lifetime value.